

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF PHYSICS
FINAL EXAMINATION

PHYS3114 – Electrodynamics
Term 3, 2021

1. Time allowed – 2 hours {+ 15 min for download/reading + 30 min for capture/upload}
2. Total number of questions – 4
3. Total marks available – 60 (pro rata to 60% of course assessment)
4. You are allowed to use your textbook and notes during the exam.
5. Prepare a single file (pdf files preferred) with all of your responses to upload to Moodle by the end of the allotted exam time.
6. Additional time will be allocated for submitting your answers
7. Your submitted file should have your name and zID on the first page of your submitted file (preferably in the header or footer on every page).
8. When uploading please use the following naming convention: PHYS3114 zID SURNAME NAME (e.g., PHYS3114 z5556789 Stark Anthony). NOTE: failure to name your exam file correctly may result in marking delays if your file is misplaced.
9. **Candidates must verify the upload is successful and contact the exam convener immediately if there is an issue.**
10. Answer all questions concisely and legibly.
11. In some cases, sketching your answer can be helpful.
12. Questions are not all worth the same marks; questions are worth the marks shown.
13. You may use a university-approved calculator.

PHYS3114 — Definitions and Formulae

$$F(s) = \int_{-\infty}^{+\infty} f(x) e^{-2\pi i x s} dx \quad (1)$$

$$\mathcal{F}\{f(x)\} = 2 \int_0^\infty f_e(x) \cos(2\pi s x) dx - 2i \int_0^\infty f_o(x) \sin(2\pi s x) dx \quad (2)$$

$$\gamma(P_1, P_2) = \frac{\iint_{\text{source}} I(\zeta, \eta) e^{\frac{-2\pi i}{\lambda}(p\zeta + q\eta)} d\zeta d\eta}{\iint_{\text{source}} I(\zeta, \eta) d\zeta d\eta} \quad (3)$$

$$V(x', y') = C_1 C_2 \lambda^2 f_o^2 \iint_{\text{bfp}} F\left(\frac{\zeta}{\lambda f_o}, \frac{\eta}{\lambda f_o}\right) e^{-i2\pi(\frac{f_o}{D}x' \frac{\zeta}{\lambda f_o} + \frac{f_o}{D}y' \frac{\eta}{\lambda f_o})} d\left(\frac{\zeta}{\lambda f_o}\right) d\left(\frac{\eta}{\lambda f_o}\right) \quad (4)$$

$$\mathcal{F}\left\{\frac{1}{2}\delta(r-a)\right\} = \pi a J_0(2\pi a \rho) \quad (5)$$

$$\mathcal{F}\left\{\prod\left(\frac{r}{2a}\right)\right\} = a \frac{J_1(2\pi a \rho)}{\rho} \quad (6)$$

$$F(\rho) = 2\pi \int_0^\infty f(r) J_0(2\pi \rho r) r dr \quad (7)$$

$$\frac{m\lambda}{2} = \frac{r_m^2}{2} \left(\frac{1}{r_0} + \frac{1}{b_0} \right) \quad (8)$$

$$\frac{d^2 P}{dt^2} + \gamma \frac{dP}{dt} + \omega_0^2 P = \frac{e^2 N}{m} \left(E + \frac{\nu P}{\epsilon_0} \right). \quad (9)$$

$$\hat{K} - 1 = \frac{\omega_p^2(\omega_{0'}^2 - \omega^2)}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} + i \frac{\omega_p^2 \omega \gamma}{(\omega_{0'}^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (10)$$

Bessel function nodes

$J_0(x)=0$, $x = 2.405, 5.5201, 8.6537, \dots$

$J_1(x)=0$, $x = 3.833, 7.0156, 10.1735, \dots$

Theorem	$f(x)$	$F(s)$
Similarity	$f(ax)$	$\frac{1}{ a }F\left(\frac{s}{a}\right)$
Linearity	$\alpha \cdot f(x) + \beta \cdot g(x)$	$\alpha \cdot F(s) + \beta \cdot G(s)$
Shift	$f(x - a)$	$e^{-2\pi ias}F(s)$
Convolution	$f(x) * g(x)$	$F(s)G(s)$
Derivative	$\frac{d}{dx}(f(x))$	$2\pi isF(s)$
Rayleigh	$\int_{-\infty}^{+\infty} f(x) ^2 dx$	$\int_{-\infty}^{+\infty} F(s) ^2 ds$

Table 1: Basic theorems of Fourier theory

PHYS3114 — (more) Definitions and Formulae

Gradient	$\nabla f = \frac{\partial f}{\partial x} \hat{x} + \frac{\partial f}{\partial y} \hat{y} + \frac{\partial f}{\partial z} \hat{z}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}$ $\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}$
Laplacian	$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$ $\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}$ $\nabla^2 f = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}$
Divergence	$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$ $\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$ $\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
Curl	$\nabla \times \mathbf{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z}$ $\nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \hat{r}$ $+ \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{\phi}$ $\nabla \times \mathbf{A} = \left[\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\phi} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right] \hat{z}$
Identities	$\nabla \times (\nabla f) = 0$ $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
Gradient Theorem	$\int_a^b (\nabla f) \cdot d\mathbf{l} = f(\mathbf{b}) - f(\mathbf{a})$
Divergence Theorem	$\oint_S \mathbf{A} \cdot d\mathbf{a} = \int_V \nabla \cdot \mathbf{A} dv$
Stokes' Theorem	$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{a}$

Coulomb's Law	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$
Biot-Savart Law	$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{I d\mathbf{l} \times \hat{r}}{r^2}$
Lorentz Force Law	$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
Maxwell's Equations	$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ $\nabla \cdot \mathbf{B} = 0 \quad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$
Electric Potential	$\mathbf{E}(\mathbf{r}) = -\nabla \phi(\mathbf{r}) - \frac{\partial \mathbf{A}}{\partial t}$
Vector Potential	$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \int_V \frac{\mathbf{J}}{r} dv$
Charge Conservation	$\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} = 0$
Current Density	$\mathbf{J} = \rho \mathbf{v} = \rho \mu \mathbf{E} = \sigma \mathbf{E}$
Poynting Vector	$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$
Stored Energy	$W = \frac{1}{2} \sum_i q_i V(\mathbf{r}_i) = \frac{1}{2} \int V(\mathbf{r}) \rho(\mathbf{r}) d^3 \mathbf{r}$ $W = \frac{1}{2} \int \left(\epsilon E^2 + \frac{B^2}{\mu} \right) dv \quad W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} dv$
Faraday's Law	$\int \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt} \quad \phi = \int_S \mathbf{B} \cdot d\mathbf{a}$
Electric Displacement	$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \quad \int_S \mathbf{D} \cdot d\mathbf{s} = Q_f$
Magnetic Field Intensity	$\mathbf{H} = \frac{1}{\mu_0} \mathbf{B} - \mathbf{M} \quad \int_L \mathbf{H} \times d\mathbf{l} = I_f$
Bound Charge Densities	$\rho_b = -\nabla \cdot \mathbf{P} \quad \sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$
Magnetization Current Densities	$\mathbf{J}_b = \nabla \times \mathbf{M} \quad \mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$
Linear Media	$\mathbf{P} = \chi_e \epsilon_0 \mathbf{E} \quad \mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon \mathbf{E}$ $\mathbf{M} = \chi_m \mathbf{H} \quad \mathbf{B} = \mu_0 (1 + \chi_m) \mathbf{H} = \mu \mathbf{H}$
Capacitance	$C = \frac{Q}{V} \quad W = \frac{1}{2} CV^2$
Inductance	$\varepsilon = -L \frac{dI}{dt} \quad W = \frac{1}{2} LI^2$

Physical Constants

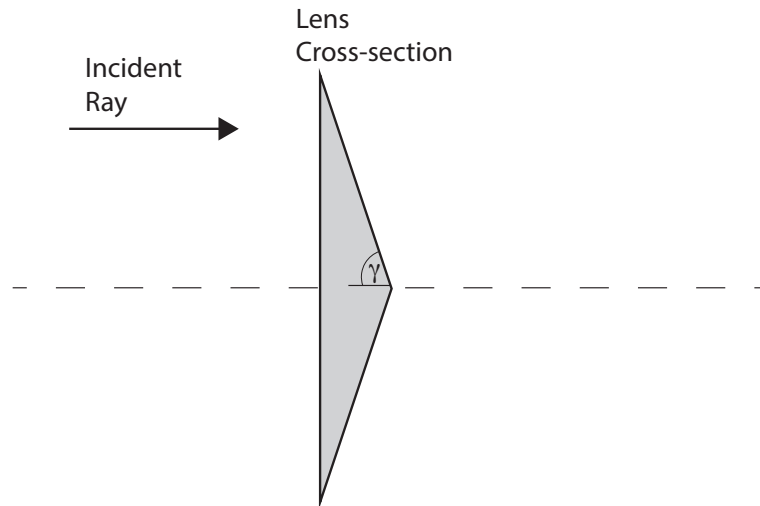
Avagadro's Number	N_A	$6.02 \times 10^{23} \text{ mole}^{-1}$
Permittivity of free space	ϵ_0	$8.85 \times 10^{-12} \text{ Fm}^{-1}$
	$\frac{1}{4\pi\epsilon_0}$	$9.0 \times 10^9 \text{ Nm}^2\text{C}^{-2}$
Permeability of free space	μ_0	$4\pi \times 10^{-7} \text{ Hm}^{-1}$
Electron charge	e	$1.6 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67 \times 10^{-27} \text{ kg}$
Speed of light in vacuum	c	$3.00 \times 10^8 \text{ ms}^{-1}$

Useful Mathematical Values

Bessel function nodes	$J_0(x) = 0$	$x = 2.405, 5.5201, 8.6537, \dots$
Bessel function nodes	$J_1(x) = 0$	$x = 3.833, 7.0156, 10.1735, \dots$

Question 1 (15 marks)

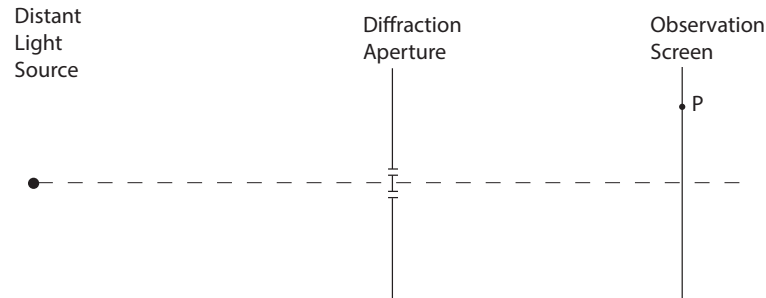
Consider a geometric optics problem consisting of paraxial rays incident on a cone shaped lens, as shown in the accompanying figure (not to scale). The apex of the cone has a half angle of γ which is close to 90° . The refractive index of the lens is, n . For this problem we can assume the thin lens condition holds.



- (a) Derive an expression for the angle that an incident ray travelling parallel to the optical axis will exit the lens. The angle should be defined with respect to the optical axis.
- (b) Using the above or otherwise, write down a ray tracing matrix that could be used to represent the action of this lens.
- (c) Derive an expression for the focus in terms of the height of the incoming ray.
- (d) Describe how the cone lens differs in behaviour from a normal thin lens.
- (e) Light passing through such a lens will generate a far field diffraction pattern that appears as an annulus. Use this information and your knowledge of diffraction theory to determine an expression for the field at the focus.

Question 2 (15 marks)

Consider a diffraction experiment conducted using a distant light source that is quasi-monochromatic. Light from the source is allowed to diffract from an ideal double slit with slit separation, d . The observation screen is placed a distance that is sufficiently large that it satisfies the far-field diffraction conditions.



- (a) Derive an expression for the **time-averaged** intensity of the field measured at point P on the observation screen in terms of the fields emitted at each of the apertures. Clearly explain all steps and approximations made.
- (b) The light source consists of a single spectral peak centred at λ_0 with a finite Gaussian width of $\delta\lambda$. Derive an expression for the time averaged intensity measured on the observation screen for this source and sketch the results, marking all important features.
- (c) A second point source with identical spectral properties is brought close to the first, where the angle subtended by the two sources is given by $\delta\theta$. Describe qualitatively how the observed diffraction pattern will change.
- (d) Comment on the relevance of this experiment to spatial and temporal coherence.

Question 3 (15 marks)

Radiation of a magnetic dipole. A time-dependent magnetic dipole is created by driving an arbitrary current $I(t)$ in a wire loop of radius b lying in the x - y plane as described in the lectures.

The magnetic dipole moment is given by

$$\mathbf{m}(t) = \pi b^2 I(t) \hat{\mathbf{z}}.$$

In the Lorentz gauge, $\nabla \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$, the electrodynamic potentials can be expressed as

$$\begin{aligned} V(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t_r)}{|\mathbf{r} - \mathbf{r}'|} d^3r', \\ \mathbf{A}(\mathbf{r}, t) &= \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t_r)}{|\mathbf{r} - \mathbf{r}'|} d^3r', \end{aligned}$$

where $t_r = t - |\mathbf{r} - \mathbf{r}'|/c$ is the retarded time.

- (a) Determine the scalar potential $V(\mathbf{r}, t)$ for the dipole.
- (b) Under some approximations, the vector potential for this system can be derived as

$$\mathbf{A} = \frac{\mu_0 b^2 \sin \theta}{4r} \left[\frac{I(t - r/c)}{r} + \frac{\dot{I}(t - r/c)}{c} \right] \hat{\phi}.$$

What approximations have been made to obtain this form of the vector potential?

- (c) Calculate the electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$, based on your answers to parts (a) and (b). Express your answers in terms of the magnetic dipole moment $\mathbf{m}(t)$.
- (d) In the static dipole limit, where $\mathbf{m}$ is constant in time, what are the electric and magnetic fields? Are they what you expect? Explain.
- (e) Returning to the general time-dependent dipole $\mathbf{m}(t)$, use the electric and magnetic fields you calculated in part (c) to find the Poynting vector $\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$ in the limit $r \rightarrow \infty$. In what approximation is r sufficiently large for this problem?
- (f) Integrating S over a large sphere, we find that the total power radiated by the dipole is

$$P = \frac{\mu_0 \ddot{\mathbf{m}}^2}{6\pi c^3}.$$

Now let us assume that we are driving an oscillating current $I(t) = I_0 \cos \omega t$ in the loop. What is the total power radiated? Express P in terms of I_0 and ω and describe briefly the physical meaning of $P(\omega)$ dependence.

Question 4 (15 marks)

A uniformly charged rod of length L and charge density λ (in lab frame) slides along the x -axis with constant speed v . At time $t = 0$ the back end passes the origin, so its position as a function of time is $x = vt$, while the front end is at $x = vt + L$.

- (a) Determine the retarded time t_1 for the back end and the corresponding retarded position x_1 .
- (b) Determine the retarded time t_2 for the front end and the corresponding retarded position x_2 .
- (c) Find the retarded scalar potential (in Lorentz gauge) at the origin as a function of time for $t > 0$.
- (d) Is your answer consistent with the Lienard-Wiechert potential in the point charge limit ($L \ll vt$, with $\lambda L = q$)? Do not assume $v \ll c$.

– End of Exam –